



Semester: Jan 2024 – April 2024		
Maximum Marks: 100	Examination: ESE Examination (KT)	Duration: 3 Hrs.
Programme code: 03		
Programme: B. Tech Electronics and Telecommunication Engineering	Class: SY	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Electronics and Telecommunication Engineering	
Course Code: 116U03C301	Name of the Course: Mathematics For Communication Engineering-I	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

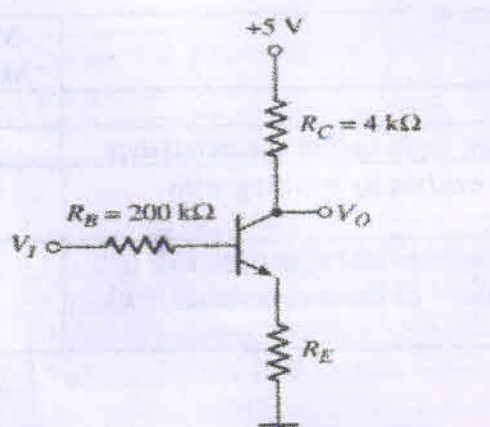
Que. No.	Question	Max. Marks
Q1	Solve any Four of the following.	20
i)	Find $L(2t^3 + \cos 4t + e^{-2t})$	5
ii)	Find $L^{-1} \left[\frac{4s+15}{16s^2-25} \right]$	5
iii)	Find Fourier series of $f(x) = 2, -1 < x < 1$	5
iv)	Prove that $4 J_n''(x) = J_{n-2}(x) - 2J_n(x) + J_{n+2}(x)$	5
v)	Prove that the points $(2, 1, 1), (0, 1, -3), (3, 2, -1)$ and $(7, 2, 7)$ are coplanar	5
vi)	Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$ along the curve $x = t, y = t^2, z = t^3$ from $t = 0$ to $t = 1$	5
Q2 A	Solve the following.	10
i)	Evaluate $\int_0^\infty e^{-t} [t^2 H(t-2)] dt$ Using Laplace transform	5
ii)	Find $L^{-1} \left[\frac{16}{(s-2)(s+2)} \right]$ using Convolution Theorem	5
OR		
Q2 A	Find the Fourier expansion of $f(x) = 4 - x^2, 0 \leq x \leq 2$ whose period is 2	10
Q 2 B	Solve any One of the following.	10
i)	Using Laplace Transform solve $(D^2 - 3D + 2)y = 4e^{2t}, y(0) = -3, y'(0) = 5$	10
ii)	Find Fourier series for $f(x) = 2x$ in $(0, 2\pi)$ Also find a_4 and b_{10}	10

Q3	Solve any Two of the following.	20
i)	(a) Obtain Complex form of Fourier series for $f(x) = e^{-2x}$ in $(-2, 2)$	05
	(b) Prove that $6J_3(x) = xJ_2(x) + xJ_4(x)$	05
ii)	Find the angle between the surfaces $x^2 + y^2 + z^2 = 12$ and $x^2 + y^2 - z = 6$ at the point $(2, -2, 2)$	10
iii)	Show that $\vec{F} = (x + 2y + 4z)\vec{i} + (2x - 3y - z)\vec{j} - (4x - y + 2z)\vec{k}$ is irrotational and find the scalar potential for $\vec{F}$ and evaluate $\int \vec{F} \cdot d\vec{r}$ along the line joining the points $(1, 2, -4)$ and $(3, 3, 2)$	10
Q4	Solve any Two of the following.	20
i)	(a) Obtain half-range sine series for $f(x) = x^3$ for $0 \leq x \leq \pi$	5
	(b) Prove that: $\int x^3 J_0(x) dx = x^3 J_1(x) - 2x^2 J_2(x) + c$	5
ii)	If $\vec{F} = x^2\vec{i} + xz\vec{j} + yz\vec{k}$ and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ find $\text{div}(\vec{F} \times \vec{r})$ and $\text{curl}(\vec{F} \times \vec{r})$	10
iii)	Verify Green's theorem in the plane for $\oint 8y^2 dx - 6xy dy$ around the boundary of the region defined by $x = 0, y = 0, x + y = 1$.	10
Q5	Solve any Four of the following.	20
i)	Find $L[t \cos^2 t]$	5
ii)	Find $L^{-1} \left[\frac{3s+1}{(s+1)(s+2)} \right]$ using partial fraction method	5
iii)	Obtain half-range cosine series for $f(x) = c - x$ for $0 \leq x \leq c$	5
iv)	Prove that $\frac{d}{dx} [x^n J_n(ax)] = ax^n J_{n-1}(ax)$	5
v)	Find the directional derivative of $\phi = 4x - 3y^2z$ at $(2, -1, 1)$ in the direction from this point towards the point $(4, -4, 8)$	5
vi)	Use Gauss's divergence theorem to evaluate $\iint_S \vec{N} \cdot \vec{F} ds$ where $\vec{F} = 4x\vec{i} + 3y\vec{j} - 7z\vec{k}$ and S is the surface bounded by $x = 0, y = 0, z = 0$ and $2x + 2y + z = 4$	5



Semester: January 2024 –April 2024		
Maximum Marks: 100	Examination: ESE Examination –(KT)	Duration:3 Hrs.
Programme code: 03 Programme: B.Tech	Class: SY	Semester: III(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC
Course Code: 116U03C302	Name of the Course: Basic Electronic Circuits	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

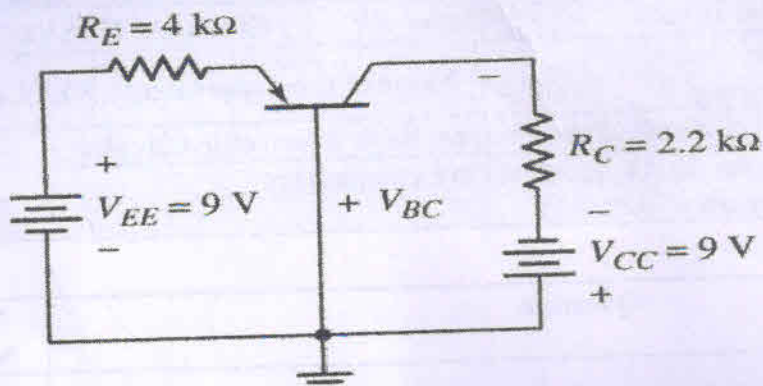
Que. No.	Question	Max. Marks
Q1	Attempt any Four	20
i)	Draw the circuit diagram and forward and reverse bias characteristics of a pn junction diode. will the characteristics change for temperature variation.	5
ii)	Draw the circuit diagram and voltage transfer characteristics of a series clipper circuit. what happens if diode is not ideal.	5
iii)	Draw the equivalent circuit diagram of a JFET when $V_G = 0V$, $V_G < V_P$, $V_G = V_P$ and $V_G > V_P$ adhere V_G is the gate voltage and V_P is the pinchoff voltage	5
iv)	Give comparison between CE and CB configuration	5
v)	Define DC loadline, Stability and Q-point	5
vi)	Draw the output characteristics of CE amplifier clearly showing the saturation region, cutoff region and active region	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	for an n-channel MOSFET biased in the saturation region, the parameters are $k_n = 50\text{mA/V}^2$, $V_{TN} = 0.6\text{ V}$, and $I_{DQ} = 0.8\text{ mA}$. Determine the V_{DSsat} and V_{GS} . Also state whether the transistor is in saturation if $V_{DS} = 5V$	5
ii)	The transistor in figure below has $\beta = 20$ determine V_I that produces $V_O = 3V$ for i) $R_E = 0\text{ K}$ ii) $R_E = 2\text{ K}$ 	5
OR		

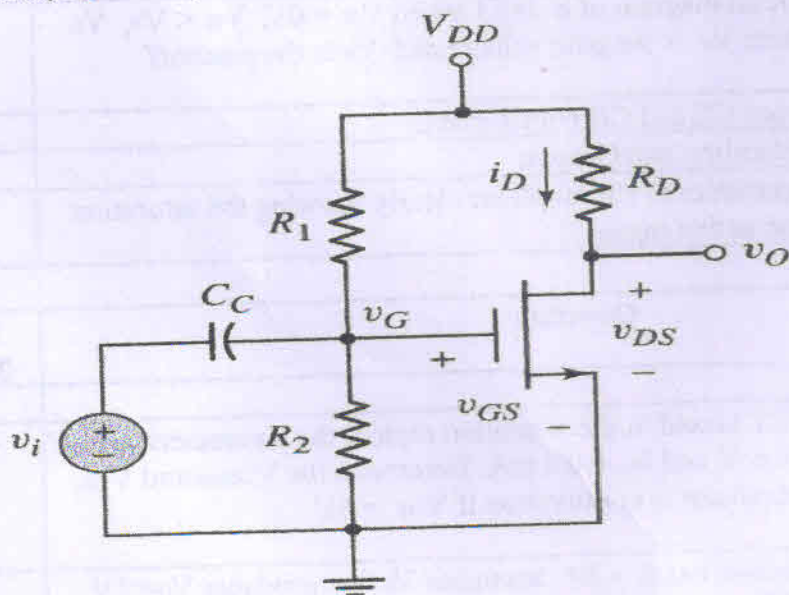
Q2 A	Give comparison between BJT and FET	10
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Q 2 B	Solve any One	10
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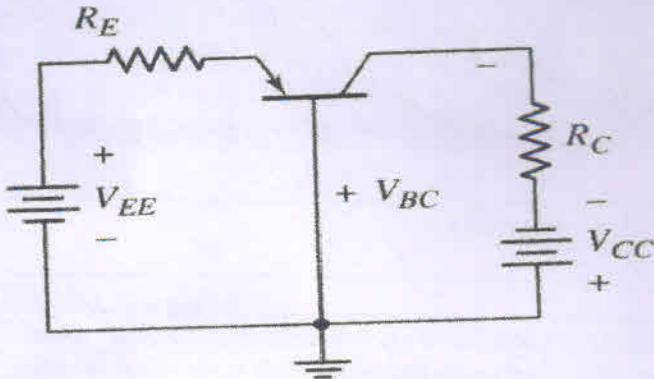
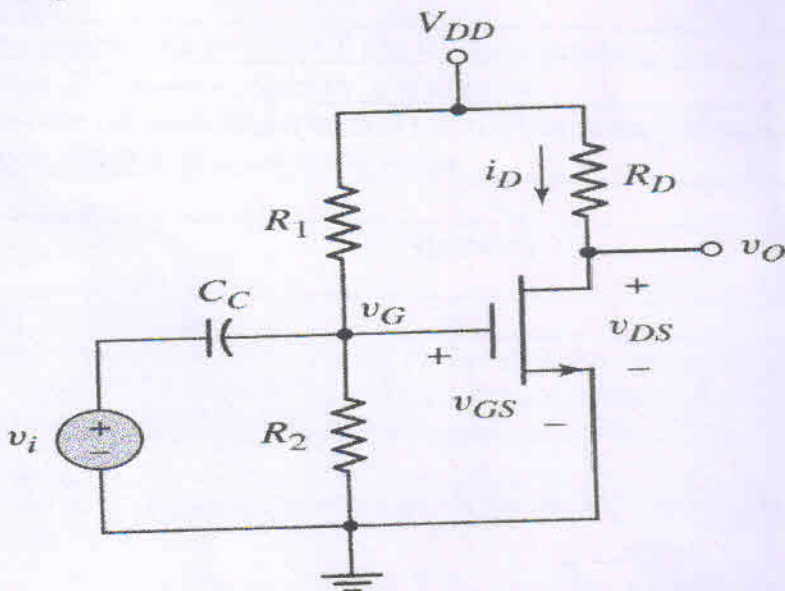
i)	For the circuit shown Determine the I_E , I_C and I_B if $\alpha = 0.91$	10
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ii)	Calculate the drain current and drain-to-source voltage of a common source circuit with an n-channel enhancement-mode MOSFET. Find the power dissipated in the transistor. For the circuit shown in the Figure. assume that $R_1 = 30\text{ k}$, $R_2 = 20\text{ k}$, $R_D = 20\text{ k}$, $V_{DD} = 10\text{ V}$, $V_{TN} = 1\text{ V}$, and $K_n = 0.1\text{ mA/V}^2$	10
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain the concept of amplification in BJT using input output characteristics	10
ii)	Draw an emitter bias circuit for a BJT amplifier. explain its working with relevant equations.	10
iii)	Explain the following terminology with a neat diagrams and equations a) Midband gain b) Cut-off frequency c) Bandwidth d) input impedance e) output impedance	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Draw small signal equivalent hybrid-pi model for mid-frequency analysis of the following circuit and derive equations for voltage gain, input resistance and output resistance 	10
ii)	Draw small signal equivalent hybrid-pi model for mid-frequency analysis of the following circuit and derive equations for voltage gain, input resistance and output resistance 	10
iii)	Define the following a) g_m b) r_d c) pinch-off voltage d) μ e) I_{DSS} with respect to FET	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Compare varactor diode with pn junction diode	5
ii)	What is a clamper circuit?. explain	5
iii)	voltage divider bias	5
iv)	Schockley equation	5
v)	Small signal hybrid pi model	5
vi)	MOS capacitor	5

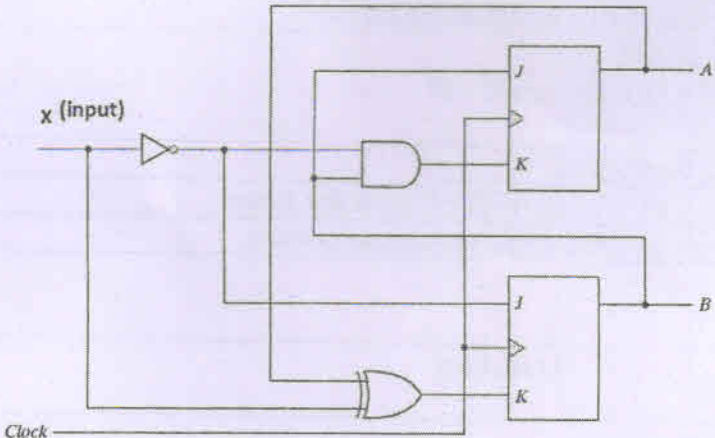


Semester: January 2024 –April 2024		
Maximum Marks: 100	Examination: ESE Examination - KT	Duration:3 Hrs.
Programme code: 03	Class: SY	Semester: _III_(SVU 2020)
Programme: BTech EXTC	BTech EXTC	
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC
Course Code: 116U03C303	Name of the Course: Digital Logic Design	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Perform the following operations. Show your calculations 1. Convert $(A20)_{16}$ into decimal. 2. $(543)_{10}$ into BCD	5
ii)	Evaluate $(78)_{10} - (53)_{10}$ using two's complement method	5
iii)	Prove the expression $(AB + C + D)(\bar{C} + D)(\bar{C} + D + E) = ABC + D$	5
iv)	Design Half adder using basic gates.	5
v)	Draw JK flip flop using gates. Write truth table for the same.	5
vi)	Differentiate between combinational and sequential circuits.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Design a full adder using 3 to 8 line decoder.	5
ii)	Write truth table for a 2 bit comparator.	5
OR		
Q2 A	If $Y(A,B,C,D) = ABC + D$ 1. Draw the corresponding K map represented by above expression 2. Write canonical SOP form of expression 3. Realise using NAND gates only	10
Q2 B	Solve any One	10
i)	Use Quine – Mc Cluskey to minimize the following expression $F(ABCD) = \sum m(0, 2, 4, 6, 7, 8, 9, 10, 13)$	10
ii)	If $Y(A, B, C, D) = \sum m(0, 1, 4, 5, 8, 9, 10, 12, 13, 14)$ I. Express the above expression in minimized the SOP form using K-Maps II. Draw the logic diagram for the above expression using only NAND gates	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Design a mod 6 synchronous counter using JK flip flops	10
ii)	Design following function using suitable size PAL(Programmable Array Logic) $F1 = \sum m(0, 2, 4, 5, 9, 11)$ $F2 = \sum m(4, 5, 13, 14, 15)$	10
iii)	Write truth tables and excitation tables for SR, JK and D flip flops	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Draw and explain 3 bit left shift register using D flip flops. Write the truth table assuming input sequence of 1, 1, 0, 1, 0, 1, 0	10
ii)	Referring to the diagram of synchronous sequential circuit below analyse the circuit. Write PS-NS table and draw state diagram.	10
		
iii)	Design mod 5 Asynchronous up counter using T flip flops	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Write short note on weighted and non-weighted codes	5
ii)	In radix (base) 5 number system a number is expressed as (413). What is its decimal and binary equivalent?	5
iii)	Write short note on Dual slope ADC	5
iv)	Compare TTL and CMOS logic families (4-5 points)	5
v)	$F = \sum m(0, 2, 4, 5)$. Realise using 8:1 mux	5
vi)	Compare synchronous and asynchronous counter.	5



Semester: January 2024 –April 2024		
Maximum Marks: 100	Examination: ESE Examination - KT	Duration:3 Hrs.
Programme code: 03	Class: SY	Semester: III(SVU 2020)
Programme: B.TECH EXTC ENGG.		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Electronics and Telecommunication Engineering
Course Code: 116U03C 304	Name of the Course: Microprocessor and Microcontroller	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Discuss segment registers of 8086.	5
ii)	Enlist and explain control flags of 8086.	5
iii)	Discuss any five features of 8051 microcontroller.	5
iv)	Explain the below pins of 8051 microcontroller. a) ALE b) RXD c) Reset	5
v)	Explain the function of following instructions of 8051. a) ANL b) DIV c) INC	5
vi)	What is the function of buses in microprocessor? Explain the function of address bus and data bus.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Draw 8086 minimum mode and explain active pins of 8086 in minimum mode.	5
ii)	What is pipelining? How it is executed in 8086 explain with suitable points.	5
	OR	
Q2 A	Explain memory banking in 8086. How to calculate physical address from segment register and pointer register? Explain with suitable example.	10
Q 2 B	Solve any One	10
i)	Write 8051 based assembly language program with comments for block transfer of 10 bytes from memory location 20h to 30h.	10
ii)	Write an 8086 based assembly language program with comments for finding smallest number in an array of n numbers.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain below instructions of 8051 with suitable example. a) XCHG A, @RP b) MOVX A, @DPTR c) MUL AB d) CALL	10
ii)	Assume XTAL = 11.0592 MHz, write a program to generate a square wave of 50 kHz frequency on port pin P2.3.	10
iii)	Interface DAC with 8051 and explain the function of DAC with suitable diagram.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	What is the function of SCON register? Explain the function of below mentioned SCON register bits. a) SM0 and SM1 b) REN c) TI d)RI	10
ii)	What is the use of timers in 8051 .Discuss the function TCON register and draw its format.	10
iii)	Discuss the necessity of ports in 8051? Explain port 3 of 8051.	10

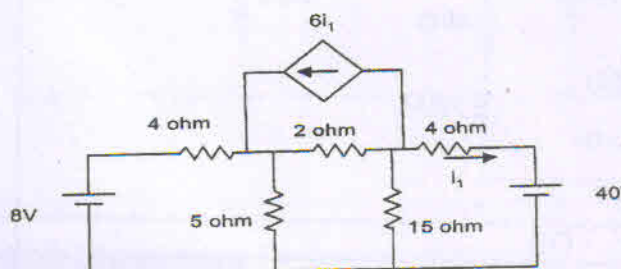
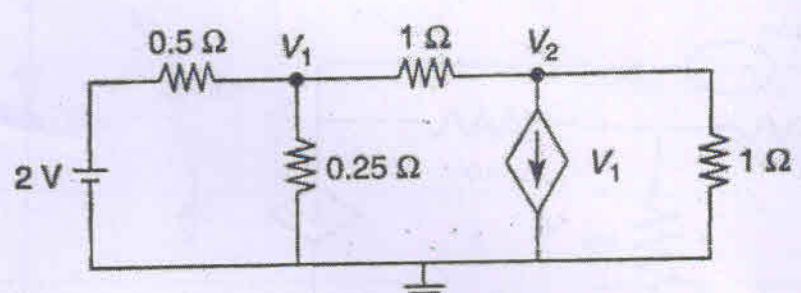
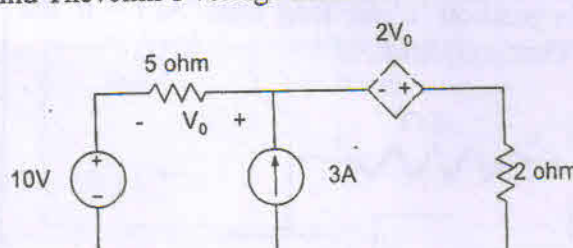
Que. No.	Question	Max. Marks
Q5	Write notes on any four	20
i)	Explain 8086 based interrupt instructions.	5
ii)	Explain any two ROTATE instructions of 8051 with suitable example	5
iii)	Write a note on timer mode3.	5
iv)	Draw an interfacing diagram of LCD with 8051 and explain in brief LCD pin functions.	5
v)	Explain register banks of 8051.	5
vi)	For 8051 system of 11.0592 MHz, find how long it takes to execute each instruction given below. a) MOV R3,# 65 b) DEC R4 c) LJMP d) NOP	5



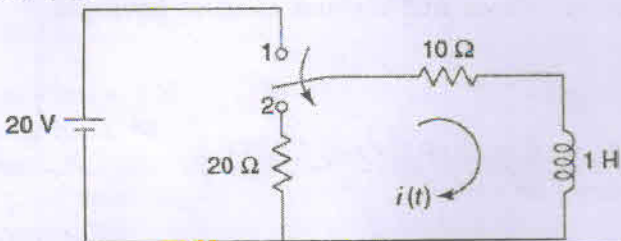
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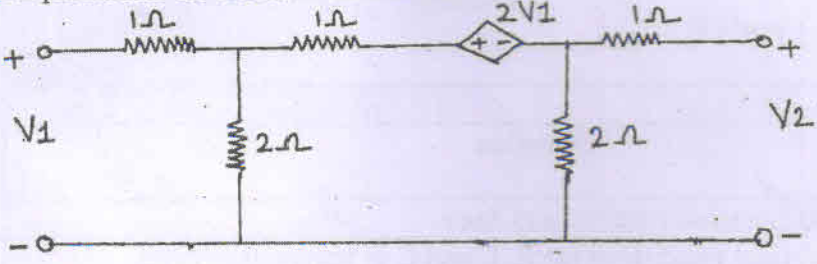
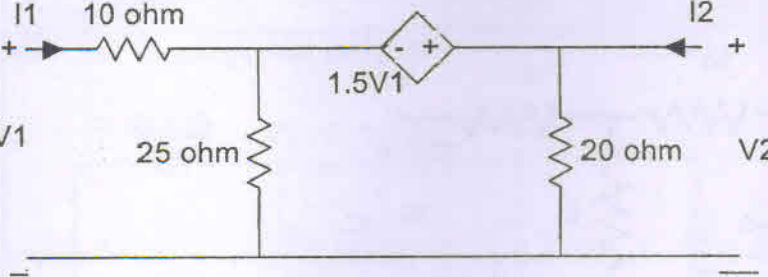
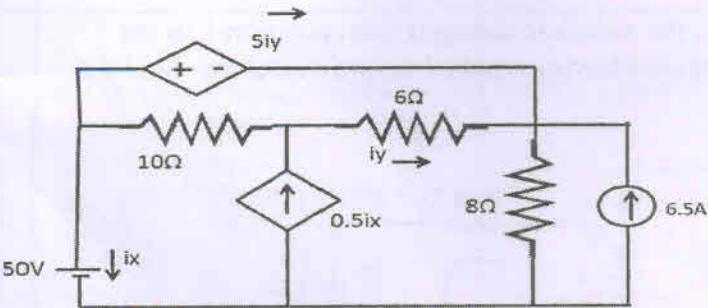
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Semester: January 2024 –April 2024		
Maximum Marks: 100	Examination: ESE Examination - KT	Duration:3 Hrs.
Programme code: 03	Class: SY	Semester:III(SVU 2020)
Programme: B.Tech in EXTC Engg.		
Name of the Constituent College:		Name of the department: EXTC
K. J. Somaiya College of Engineering		
Course Code: 116U03C305	Name of the Course: Electrical Network Theory	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

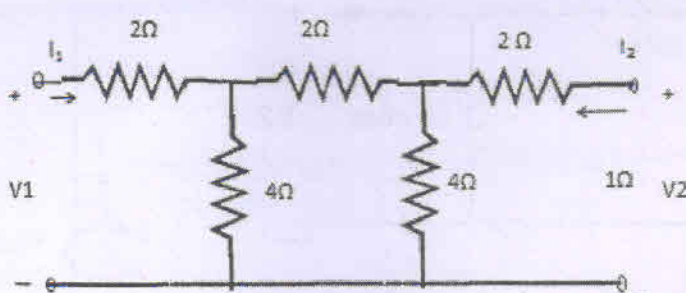
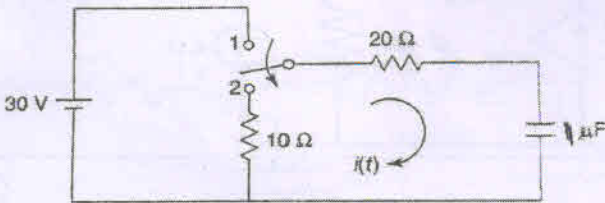
Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Find current through 2 ohm using KVL.	5
		
ii)	Find V_1 and V_2 using nodal analysis	5
		
iii)	Find Thevenin's voltage across load resistance 2 ohm in the given circuit	5
		
iv)	Realize Foster I form of the LC impedance function. $Z(s) = \frac{(s^2 + 1)(s^2 + 3)}{s(s^2 + 2)}$	5
v)	Write reciprocity and symmetry condition for Z parameter	5
vi)	Write note on types of filters	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	For the circuit shown determine thevenin's voltage across terminals a-b.	5
The circuit diagram for Q2 A i) shows a 100V DC voltage source on the left. A dependent current source labeled $5i_x$ is in parallel with a 10Ω resistor. The current through the 10Ω resistor is labeled i_x. A 4A DC current source is in parallel with the dependent source and resistor. To the right of the 4A source is a 5Ω resistor connected to terminals A and B. The voltage across terminals A and B is the Thevenin voltage.		5
ii)	Find the thevenin's equivalent resistance at terminals a-b in circuit shown below.	5
The circuit diagram for Q2 A ii) shows a $40\angle 90^\circ$ V voltage source in series with a 5Ω resistor. This is followed by a parallel combination of an 8Ω resistor and a $-j2\Omega$ capacitor. Then, a $3\angle 0^\circ$ A current source is in parallel with a 10Ω resistor and a $j4\Omega$ inductor. Finally, there is a series combination of a 20Ω resistor and a $j15\Omega$ inductor connected to terminals a and b. The current entering terminal a is labeled I_o.		5
OR		
Q2 A	For the circuit shown determine current flowing through 6Ω resistance using Superposition theorem	10
The circuit diagram for Q2 A OR shows a 20V DC voltage source in series with a 2Ω resistor. This is followed by a parallel combination of a 10Ω resistor and a 4Ω resistor. A 6Ω resistor is connected in parallel across the output terminals, with voltage V_x across it. A dependent current source labeled $\frac{V_x}{2}$ is also in parallel. A 10A DC current source is in parallel with the 2Ω resistor.		10
Q2 B	Solve any One	10
i)	In the network shown the switch is in position 'a' for long time. At $t = 0$, the switch is moved from position a to b. Find $v_c(t)$ for $t > 0$.	10
The circuit diagram for Q2 B i) shows a 10V DC voltage source in series with a 1Ω resistor. A switch is connected to terminal 'a' for $t < 0$ and terminal 'b' for $t > 0$. Terminal 'a' is connected to a $6F$ capacitor. Terminal 'b' is connected to a 1Ω resistor in series with a $3F$ capacitor. The voltage across the $3F$ capacitor is labeled $V_c(t)$. The current through the $6F$ capacitor is labeled $i(t)$.		10

ii)	In the network shown below, the switch is initially at position 1 for a long time. At $t = 0$, the switch is changed to position 2. Find current $i(t)$ for $t > 0$. 	10
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Find Z parameter for the network shown: 	10
ii)	Find ABCD parameters for the network shown 	10
iii)	Use mesh analysis and find current through 8Ω. 	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Each of the two series element of a T section low pass filter consists of an inductor of 60 mH having negligible resistance and a shunt element having a capacitance of 0.2 μ F. Calculate a) Cut-off frequency b) Nominal impedance c) Characteristic impedance at frequency of 1KHz and 5 KHz	10
ii)	Test the given F(s) for positive real function. $F(s) = \frac{2s^3 + 2s^2 + 3s + 2}{s^2 + 1}$	10
iii)	Realize Cauer I and II forms of the following LC impedance function. $Z(s) = [3(s^2+1)] / [s(s^2+9)]$	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Tabulate initial and final conditions for R, L and C in transient circuits.	5
ii)	Write two port equations for Y and ABCD parameters.	5
iii)	Check whether $P(s) = 2s^6 + s^5 + 13s^4 + 16s^3 + 56s^2 + 25s + 25$ is Hurwitz.	5
iv)	For the two port network shown, determine Z parameters. 	5
v)	In the network shown, the switch is changed from position 1 to the position 2 at $t = 0$, steady condition having reached before switching. Find the values of i , $\frac{di}{dt}$ at $t = 0^+$. 	5
vi)	Explain self-inductance and mutual inductance.	5